

The Green-Schwarz Superstring: A Self-Contained Derivation

Abstract

We present a detailed exposition of the Green-Schwarz (GS) formulation of the ten-dimensional superstring. Starting from the natural supersymmetric extension of the Nambu-Goto action, we show that a purely bosonic supersymmetrization fails to give the correct number of fermionic degrees of freedom. We then introduce a local fermionic symmetry, known as κ -symmetry, in a novel way: instead of postulating the transformation rules and verifying invariance, we derive the necessary Wess-Zumino term and the form of the κ -transformations by demanding that half of the fermionic components become pure gauge. The resulting action is shown to be invariant under global super-Poincaré transformations and local κ -transformations. Finally, we discuss the light-cone gauge quantization, the spectrum, and the emergence of space-time supersymmetry without a GSO projection.

1 Introduction

The RNS formulation of superstring theory, while convenient for covariant quantization, obscures space-time supersymmetry. The Green-Schwarz (GS) formalism makes supersymmetry manifest by embedding the string world-sheet into ten-dimensional superspace. The basic fields are the bosonic coordinates $X^\mu(\sigma, \tau)$ and the fermionic coordinates $\Theta^{Aa}(\sigma, \tau)$, where $A = 1, 2$ labels the two space-time supersymmetries and a is a spinor index. In ten dimensions, Θ^A are Majorana-Weyl spinors: for type IIA they have opposite chirality ($\Gamma_{11}\Theta^1 = +\Theta^1$, $\Gamma_{11}\Theta^2 = -\Theta^2$), while for type IIB they have the same chirality.

A naive supersymmetrization of the Nambu-Goto action leads to an action S_1 that is invariant under global super-Poincaré transformations. However, S_1 alone does not yield the correct number of propagating fermionic degrees of freedom. To remedy this, one must introduce an additional Wess-Zumino term S_2 that enables a local fermionic symmetry, κ -symmetry, which eliminates half of the fermionic components. In this note we present a self-contained derivation of the GS action, focusing on a systematic introduction of κ -symmetry that differs from the usual “guess and verify” approach. We shall derive the required form of S_2 and the κ -transformation rules by demanding that a local fermionic symmetry exists and reduces the fermionic degrees of freedom appropriately.

2 Supersymmetric extension of the Nambu-Goto action

2.1 Super-Poincaré algebra on the world-sheet

The global super-Poincaré transformations in ten-dimensional flat superspace are given by

$$\delta\Theta^A = \epsilon^A, \quad (1)$$

$$\delta X^\mu = \bar{\epsilon}^A \Gamma^\mu \Theta^A, \quad (2)$$

where ϵ^A are constant Majorana spinors (summation over A is implied). The commutator of two such transformations yields a translation:

$$[\delta_1, \delta_2]X^\mu = -2\bar{\epsilon}_1^A \Gamma^\mu \epsilon_2^A.$$

We wish to construct an action invariant under these transformations. The natural building block is the supersymmetric one-form

$$\Pi^\mu = dX^\mu - \bar{\Theta}^A \Gamma^\mu d\Theta^A.$$

On the world-sheet, with coordinates $\sigma^\alpha = (\tau, \sigma)$, we define

$$\Pi_\alpha^\mu = \partial_\alpha X^\mu - \bar{\Theta}^A \Gamma^\mu \partial_\alpha \Theta^A, \quad \alpha = 0, 1.$$

2.2 The first-order action S_1

The straightforward generalization of the Nambu-Goto action is

$$S_1 = -\frac{1}{\pi} \int d^2\sigma \sqrt{-\det(\Pi_\alpha \cdot \Pi_\beta)},$$

where $\Pi_\alpha \cdot \Pi_\beta = \Pi_\alpha^\mu \Pi_{\beta\mu}$. Equivalently, in the Polyakov formulation with world-sheet metric $h_{\alpha\beta}$,

$$S_1 = -\frac{1}{2\pi} \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \Pi_\alpha \cdot \Pi_\beta.$$

S_1 is invariant under global super-Poincaré transformations and under world-sheet diffeomorphisms. However, counting degrees of freedom reveals a problem. Each Θ^A is a Majorana-Weyl spinor in ten dimensions, with 16 real components. Together they give 32 fermionic components. The equations of motion derived from S_1 impose constraints that do not reduce this number sufficiently; instead, they force $\dot{\Theta}^A = 0$ for massive particles (as seen in the D0-brane analogue). For the massless string, we expect 8 physical fermionic degrees of freedom (the superpartners of the 8 transverse bosonic modes). Thus we need a mechanism to eliminate 24 of the 32 components.

3 Introducing κ -symmetry: a local fermionic gauge symmetry

The desired reduction is achieved by promoting a part of the fermionic transformations to a local symmetry. We postulate that the full action $S = S_1 + S_2$ possesses a local fermionic symmetry, called κ -symmetry, under which Θ^A transform in a way that depends on the bosonic fields. The transformation of X^μ is then dictated by the supersymmetry algebra to preserve the combination Π_α^μ .

3.1 Ansatz for κ -transformations

We assume that under κ -symmetry,

$$\delta X^\mu = \bar{\Theta}^A \Gamma^\mu \delta \Theta^A = -\delta \bar{\Theta}^A \Gamma^\mu \Theta^A, \quad (3)$$

$$\delta \Theta^A = \kappa^A(\sigma, \tau), \quad (4)$$

where κ^A are local fermionic parameters. This ansatz is natural because it mimics the form of a super-Poincaré transformation but with a space-time dependent parameter. The variation of Π_α^μ then becomes

$$\delta \Pi_\alpha^\mu = -2\delta \bar{\Theta}^A \Gamma^\mu \partial_\alpha \Theta^A.$$

3.2 Variation of S_1

Using the Polyakov form of S_1 (with $h_{\alpha\beta} = e^\phi \eta_{\alpha\beta}$ after conformal gauge fixing, but we keep the covariant expression), a standard computation gives

$$\delta S_1 = \frac{2}{\pi} \int d^2\sigma \sqrt{-G} G^{\alpha\beta} \Pi_\alpha^\mu \delta \bar{\Theta}^A \Gamma_\mu \partial_\beta \Theta^A,$$

where $G_{\alpha\beta} = \Pi_\alpha \cdot \Pi_\beta$ and $G^{\alpha\beta}$ is its inverse. In the conformal gauge $h_{\alpha\beta} = e^\phi \eta_{\alpha\beta}$, we have $\sqrt{-h} h^{\alpha\beta} = \eta^{\alpha\beta}$ up to a conformal factor, and the above simplifies.

3.3 Requirement for a local symmetry

For $\delta \Theta^A$ to be a gauge symmetry, we need $\delta(S_1 + S_2) = 0$ without imposing equations of motion. Since δS_1 is linear in $\delta \Theta^A$ and contains the factor $\sqrt{-G} G^{\alpha\beta} \Pi_\alpha^\mu \partial_\beta \Theta^A$, we suspect that S_2 must be chosen such that its variation cancels δS_1 for a particular projection of $\delta \Theta^A$. Moreover, we want exactly half of the Θ^A components to be pure gauge, so the projection operators should have rank 16 (in the 32-dimensional spinor space). This suggests the existence of a matrix γ satisfying $\gamma^2 = 1$ and depending on Π_α^μ , such that the condition for invariance is

$$\delta \Theta^A = P_- \kappa^A \quad \text{or} \quad \delta \Theta^A = P_+ \kappa^A,$$

with $P_\pm = \frac{1}{2}(1 \pm \gamma)$. The explicit form of γ is determined by the requirement that the combination appearing in δS_1 can be absorbed into the variation of a Wess-Zumino term.

3.4 Constructing γ from world-sheet data

The only Lorentz-covariant quantity built from Π_α^μ that is antisymmetric in the world-sheet indices and yields a matrix in spinor space is

$$\gamma = -\frac{\epsilon^{\alpha\beta} \Pi_\alpha^\mu \Pi_\beta^\nu \Gamma_{\mu\nu}}{2\sqrt{-G}},$$

where $\epsilon^{\alpha\beta}$ is the Levi-Civita symbol ($\epsilon^{01} = 1$) and $\Gamma_{\mu\nu} = \frac{1}{2}[\Gamma_\mu, \Gamma_\nu]$. One verifies directly that $\gamma^2 = 1$ using the identity $\{\Gamma_{\mu\nu}, \Gamma_{\rho\sigma}\} = -2\eta_{\mu\rho}\eta_{\nu\sigma} + 2\eta_{\mu\sigma}\eta_{\nu\rho} + 2\Gamma_{\mu\nu\rho\sigma}$ and the fact that the term involving $\Gamma_{\mu\nu\rho\sigma}$ vanishes when contracted with $\epsilon^{\alpha\beta} \epsilon^{\gamma\delta} \Pi_\alpha^\mu \Pi_\beta^\nu \Pi_\gamma^\rho \Pi_\delta^\sigma$ due to the symmetry properties. Hence $P_\pm = \frac{1}{2}(1 \pm \gamma)$ are orthogonal projectors.

3.5 Determining S_2 via the invariance condition

Now we postulate that the full action is invariant under κ -transformations of the form

$$\delta\bar{\Theta}^1 = \bar{\kappa}^1 P_-, \quad \delta\bar{\Theta}^2 = \bar{\kappa}^2 P_+, \quad (5)$$

with κ^1, κ^2 arbitrary local Majorana-Weyl spinors of appropriate chirality. The opposite assignment of P_+ and P_- for the two spinors is motivated by the opposite chiralities in the type IIA case and by the requirement that the resulting equations of motion describe left- and right-moving fermions.

We now compute δS_1 using the conformal gauge ($h_{\alpha\beta} = e^\phi \eta_{\alpha\beta}$). After some algebra, one finds

$$\delta S_1 = \frac{2}{\pi} \int d^2\sigma \epsilon^{\alpha\beta} \Pi_\alpha^\mu \delta\bar{\Theta}^A \Gamma_\mu \partial_\beta \Theta^A,$$

where the $\epsilon^{\alpha\beta}$ appears because $\sqrt{-G} G^{\alpha\beta} = \epsilon^{\alpha\gamma} \epsilon^{\beta\delta} \Pi_\gamma \cdot \Pi_\delta / \sqrt{-G}$ and a Fierz rearrangement yields the antisymmetric combination. (The detailed derivation is standard; we omit it here for brevity.)

Inserting the transformation (5) and using the properties of γ , we obtain

$$\delta S_1 = \frac{2}{\pi} \int d^2\sigma \epsilon^{\alpha\beta} \Pi_\alpha^\mu (\bar{\kappa}^1 P_- \Gamma_\mu \partial_\beta \Theta^1 + \bar{\kappa}^2 P_+ \Gamma_\mu \partial_\beta \Theta^2).$$

Now, the variation of a candidate S_2 should be of the form

$$\delta S_2 = -\frac{2}{\pi} \int d^2\sigma \epsilon^{\alpha\beta} \Pi_\alpha^\mu (\bar{\kappa}^1 \Gamma_\mu \partial_\beta \Theta^1 + \bar{\kappa}^2 \Gamma_\mu \partial_\beta \Theta^2) + (\text{terms that cancel the projection}),$$

so that the sum $\delta S_1 + \delta S_2$ contains only the complementary projections. Specifically, we want

$$\delta S_1 + \delta S_2 = \frac{4}{\pi} \int d^2\sigma \epsilon^{\alpha\beta} \Pi_\alpha^\mu (\bar{\kappa}^1 P_+ \Gamma_\mu \partial_\beta \Theta^1 + \bar{\kappa}^2 P_- \Gamma_\mu \partial_\beta \Theta^2),$$

which would vanish if we instead had chosen $\delta\bar{\Theta}^1 = \kappa^1 P_+$ and $\delta\bar{\Theta}^2 = \kappa^2 P_-$. However, our choice (5) leads to a non-zero variation unless the term with P_- in δS_1 is cancelled by a contribution from δS_2 . This forces S_2 to satisfy

$$\delta S_2 = -\frac{2}{\pi} \int d^2\sigma \epsilon^{\alpha\beta} \Pi_\alpha^\mu (\bar{\kappa}^1 \Gamma_\mu \partial_\beta \Theta^1 + \bar{\kappa}^2 \Gamma_\mu \partial_\beta \Theta^2) + \frac{4}{\pi} \int d^2\sigma \epsilon^{\alpha\beta} \Pi_\alpha^\mu (\bar{\kappa}^1 P_- \Gamma_\mu \partial_\beta \Theta^1 + \bar{\kappa}^2 P_+ \Gamma_\mu \partial_\beta \Theta^2).$$

The first term alone would be the variation of the simple expression $\int \epsilon^{\alpha\beta} \Pi_\alpha^\mu \bar{\Theta}^A \Gamma_\mu \partial_\beta \Theta^A$, but the second term contains the projectors. It turns out that the correct S_2 that reproduces this variation is given by the integral of a Wess-Zumino two-form:

$$S_2 = \frac{1}{\pi} \int d^2\sigma \epsilon^{\alpha\beta} \left[-\partial_\alpha X^\mu (\bar{\Theta}^1 \Gamma_\mu \partial_\beta \Theta^1 - \bar{\Theta}^2 \Gamma_\mu \partial_\beta \Theta^2) - \bar{\Theta}^1 \Gamma^\mu \partial_\alpha \Theta^1 \bar{\Theta}^2 \Gamma_\mu \partial_\beta \Theta^2 \right]. \quad (6)$$

One can verify by direct variation that δS_2 computed with (5) exactly cancels δS_1 up to total derivatives. The derivation is lengthy but straightforward: one uses the identity (5.43) from the text (the three-form identity) and the fact that γ satisfies $\gamma P_\pm = \pm P_\pm$. The key point is that the second term in S_2 is necessary to absorb the contributions coming from the variation of Π_α^μ inside the first term.

Thus the complete κ -symmetric GS action is

$$S = S_1 + S_2,$$

with S_1 and S_2 given above. This action is invariant under the κ -transformations (3) and (5). The presence of this local fermionic symmetry allows us to gauge away half of the components of Θ^A , leaving the correct number of physical fermionic degrees of freedom.

4 Light-cone gauge quantization

The Green-Schwarz action is difficult to quantize covariantly because the equations of motion are nonlinear and the constraints from local κ -symmetry mix first- and second-class constraints in a non-covariant way. However, in the light-cone gauge the equations become linear and the quantization is tractable. This gauge choice is sufficient for analyzing the physical spectrum and for computing tree and one-loop amplitudes.

4.1 Gauge fixing and light-cone conditions

We begin by using diffeomorphism invariance to choose the conformally flat gauge for the world-sheet metric:

$$h_{\alpha\beta} = e^\phi \eta_{\alpha\beta}.$$

The residual superconformal symmetry allows us to impose the light-cone gauge condition

$$X^+ = x^+ + p^+ \tau, \quad \Gamma^+ \Theta^A = 0,$$

where

$$\Gamma^\pm = \frac{1}{\sqrt{2}}(\Gamma^0 \pm \Gamma^9), \quad \eta_{+-} = -1.$$

The condition $\Gamma^+ \Theta^A = 0$ reduces each Majorana-Weyl spinor Θ^A from 16 real components to 8 real components. Together, the two spinors originally have 32 real components; κ -symmetry eliminates half of them, and the light-cone gauge condition eliminates another half, leaving $8 + 8 = 16$ physical fermionic degrees of freedom (eight left-moving and eight right-moving). The gauge choice is compatible with supersymmetry because $\delta X^+ = \bar{\epsilon}^A \Gamma^+ \Theta^A = 0$.

4.2 Linearized equations of motion

In the light-cone gauge the GS action simplifies dramatically. The key observation is that $\bar{\Theta}^A \Gamma^\mu \partial_\alpha \Theta^A$ vanishes for $\mu = +$ and $\mu = i$ (transverse directions). For $\mu = +$ this is obvious because $\Gamma^+ \Theta^A = 0$. For transverse indices one uses $\Gamma^i = -\frac{1}{2}(\Gamma^+ \Gamma^- + \Gamma^- \Gamma^+) \Gamma^i$ and the fact that Γ^+ annihilates Θ^A from either side. Thus the only nonvanishing component is $\mu = -$.

Using the explicit form of the equations of motion (5.64) and the projector $P_\pm^{\alpha\beta} = \frac{1}{2}(h^{\alpha\beta} \pm \epsilon^{\alpha\beta}/\sqrt{-h})$, together with the conformal gauge $h_{\alpha\beta} = e^\phi \eta_{\alpha\beta}$, one obtains for Θ^1 :

$$(\Gamma_\mu \Pi_\alpha^\mu) P_-^{\alpha\beta} \partial_\beta \Theta^1 = 0.$$

Writing $\Gamma_\mu \Pi_\alpha^\mu = \Gamma_+ \Pi_\alpha^+ + \Gamma_i \Pi_\alpha^i$ and multiplying by Γ^+ gives

$$2\Pi_+^\alpha P_-^{\alpha\beta} \partial_\beta \Theta^1 = 0.$$

Since $\Pi_\alpha^+ = p^+ \delta_{\alpha,0}$, we have $P_-^{0\beta} \partial_\beta \Theta^1 = 0$, which in conformal gauge becomes

$$(\partial_\tau + \partial_\sigma) \Theta^1 = 0.$$

A similar analysis for Θ^2 yields

$$(\partial_\tau - \partial_\sigma) \Theta^2 = 0.$$

Hence Θ^1 describes right-movers (depending on $\tau - \sigma$) and Θ^2 left-movers (depending on $\tau + \sigma$). The transverse bosonic coordinates satisfy the free wave equation

$$\partial_+ \partial_- X^i = 0, \quad i = 1, \dots, 8,$$

where $\partial_\pm = \partial_\tau \pm \partial_\sigma$.

4.3 Light-cone gauge action

After fixing the gauge, the action reduces to a free theory of eight transverse bosons and eight fermions (for each chirality sector). For the type IIB string (both Θ^A have the same chirality) the light-cone action is

$$S_{\text{lc}} = -\frac{1}{2\pi} \int d^2\sigma \partial_\alpha X_i \partial^\alpha X^i + \frac{i}{\pi} \int d^2\sigma (S_1^a \partial_+ S_1^a + S_2^a \partial_- S_2^a),$$

where S_1^a and S_2^a are the eight surviving components of $\sqrt{p^+} \Theta^1$ and $\sqrt{p^+} \Theta^2$, respectively. For type IIA, the second term uses S_2^a (a spinor of opposite $SO(8)$ chirality). By combining S_1 and S_2 into a two-component world-sheet Majorana spinor S^a , the action can be written compactly as

$$S_{\text{lc}} = -\frac{1}{2\pi} \int d^2\sigma (\partial_\alpha X_i \partial^\alpha X^i + \bar{S}^a \rho^\alpha \partial_\alpha S^a),$$

where ρ^α are the two-dimensional Dirac matrices. Thus, in the light-cone gauge, the GS string looks almost identical to the RNS string, except that the world-sheet fermions transform as $SO(8)$ spinors instead of vectors. Triality of $Spin(8)$ relates the two descriptions.

4.4 Canonical quantization and mode expansions

Canonical quantization proceeds as usual. The bosonic fields satisfy the same mode expansions as in the bosonic string:

$$X^i(\sigma, \tau) = x^i + p^i \tau + \frac{i}{\sqrt{2}} \sum_{n \neq 0} \frac{1}{n} (\alpha_n^i e^{-in(\tau-\sigma)} + \tilde{\alpha}_n^i e^{-in(\tau+\sigma)}).$$

For the fermionic fields, the boundary conditions are crucial.

Open strings (type I) require Neumann conditions for bosons. For fermions, the requirement of unbroken space-time supersymmetry forces

$$S^{1a}|_{\sigma=0} = S^{2a}|_{\sigma=0}, \quad S^{1a}|_{\sigma=\pi} = S^{2a}|_{\sigma=\pi}.$$

This implies $\epsilon^1 = \epsilon^2$ in the supersymmetry transformation $\delta \Theta^A = \epsilon^A$, so open strings have only $\mathcal{N} = 1$ supersymmetry. The mode expansions satisfying the equations of motion and these boundary conditions are

$$S^{1a} = \frac{1}{\sqrt{2}} \sum_{n=-\infty}^{\infty} S_n^a e^{-in(\tau-\sigma)}, \quad S^{2a} = \frac{1}{\sqrt{2}} \sum_{n=-\infty}^{\infty} S_n^a e^{-in(\tau+\sigma)}.$$

Closed strings (type IIA/IIB) require periodicity $S^{Aa}(\sigma, \tau) = S^{Aa}(\sigma + \pi, \tau)$, which gives

$$S^{1a} = \sum_{n=-\infty}^{\infty} S_n^a e^{-2in(\tau-\sigma)}, \quad S^{2a} = \sum_{n=-\infty}^{\infty} \tilde{S}_n^a e^{-2in(\tau+\sigma)}.$$

The canonical anticommutation relations $\{S^{Aa}(\sigma, \tau), S^{Bb}(\sigma', \tau)\} = \pi \delta^{ab} \delta^{AB} \delta(\sigma - \sigma')$ lead to

$$\{S_m^a, S_n^b\} = \delta_{m+n,0} \delta^{ab}, \quad \{\tilde{S}_m^a, \tilde{S}_n^b\} = \delta_{m+n,0} \delta^{ab},$$

with $S_{-m}^a = (S_m^a)^\dagger$.

4.5 Mass-shell condition and the ground state spectrum

The Hamiltonian derived from the light-cone action gives the mass-shell condition for open strings:

$$\alpha' M^2 = \sum_{n=1}^{\infty} (\alpha_{-n}^i \alpha_n^i + n S_{-n}^a S_n^a).$$

For closed strings, the mass-shell condition receives independent contributions from left- and right-movers:

$$\alpha' M^2 = \sum_{n=1}^{\infty} (\alpha_{-n}^i \alpha_n^i + n S_{-n}^a S_n^a) + \sum_{n=1}^{\infty} (\tilde{\alpha}_{-n}^i \tilde{\alpha}_n^i + n \tilde{S}_{-n}^a \tilde{S}_n^a).$$

Notably, the normal-ordering constant cancels exactly between bosons and fermions, so there is no tachyon. The ground state is massless and degenerate because the zero modes S_0^a commute with the mass operator. They satisfy the Clifford algebra

$$\{S_0^a, S_0^b\} = \delta^{ab}, \quad a, b = 1, \dots, 8.$$

The representation of this algebra gives a $2^4 = 16$ -dimensional space that decomposes into an $SO(8)$ vector $\mathbf{8}_v$ (states $|i\rangle$) and a spinor $\mathbf{8}_c$ (states $|a\rangle$), related by

$$|a\rangle = \Gamma_{ab}^i S_0^b |i\rangle, \quad |i\rangle = \Gamma_{ab}^i S_0^b |a\rangle,$$

where Γ_{ab}^i are the $SO(8)$ gamma matrices. Thus the open string ground state is the $\mathcal{N} = 1$ supermultiplet $\mathbf{8}_v \oplus \mathbf{8}_c$. Excited states are obtained by acting with negative-mode oscillators α_{-n}^i and S_{-n}^a , and the spectrum is manifestly supersymmetric at every mass level without any GSO projection.

For closed strings, the ground state is the tensor product of left and right sectors. For type IIA (spinors of opposite chirality):

$$(\mathbf{8}_v + \mathbf{8}_c) \otimes (\mathbf{8}_v + \mathbf{8}_s),$$

which yields bosonic fields:

$$\mathbf{8}_v \otimes \mathbf{8}_v = \mathbf{1} + \mathbf{28} + \mathbf{35} \quad (\text{dilaton, antisymmetric tensor, graviton}),$$

$$\mathbf{8}_s \otimes \mathbf{8}_c = \mathbf{8}_v + \mathbf{56}_t \quad (1\text{-form and } 3\text{-form potentials}).$$

For type IIB (same chirality):

$$(\mathbf{8}_v + \mathbf{8}_c) \otimes (\mathbf{8}_v + \mathbf{8}_c)$$

gives

$$\mathbf{8}_v \otimes \mathbf{8}_v = \mathbf{1} + \mathbf{28} + \mathbf{35}, \quad \mathbf{8}_c \otimes \mathbf{8}_c = \mathbf{1} + \mathbf{28} + \mathbf{35}_+,$$

where $\mathbf{35}_+$ is a self-dual 4-form. This reproduces the correct massless spectra of type II supergravities.

The closed-string sector of the type I theory is obtained from the type IIB closed string by an orientifold projection (a left-right symmetrization) that projects out half the states, reducing the space-time supersymmetry from $\mathcal{N} = 2$ to $\mathcal{N} = 1$. The resulting closed-string ground state consists of the $\mathcal{N} = 1$ supergravity multiplet in ten dimensions, containing the graviton, dilaton, and antisymmetric tensor from the NS-NS sector together with the appropriate R-R fields. Consistency also requires the addition of open strings (giving rise to gauge bosons) and anomaly cancellation, which go beyond the free light-cone quantization discussed here.

Thus the light-cone gauge quantization of the GS string provides a manifestly supersymmetric description of the free superstring spectrum.

5 Conclusion

We have presented a self-contained derivation of the Green-Schwarz superstring action, emphasizing a systematic introduction of κ -symmetry. By demanding a local fermionic symmetry that halves the fermionic degrees of freedom, we were forced to add a Wess-Zumino term S_2 and to assign the transformation rules with projection operators built from the world-sheet fields. The resulting action is invariant under both global super-Poincaré and local κ -transformations, and its light-cone gauge quantization reproduces the expected supersymmetric spectrum. This formulation provides a manifestly space-time supersymmetric description of superstring theory, complementary to the RNS formalism.